Beyond Keywords: A Clinician-Reviewed Synthetic Benchmark for Stigmatizing Language Detection in Substance Use Disorder

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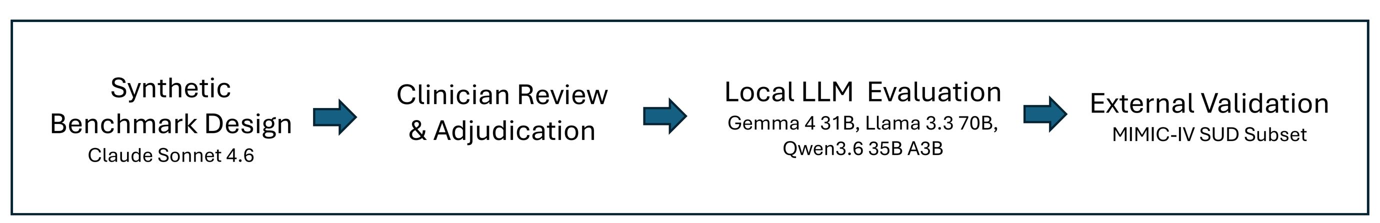
INTRODUCTION

- Stigmatizing language in electronic health records (EHRs) is associated with bias and may negatively influence patient experience and engagement in substance use disorder (SUD) care.[1]
- Traditional detection approaches have relied on manual review or natural language processing of annotated clinical notes, creating challenges for scalability and privacy.[2,3]
- Large language models show promise for automated stigma detection[4,5] and documentation auditing,[6] but contextual ambiguity involving attribution, clinical hedging, and nuanced framing remains challenging.
- We developed a clinician-reviewed synthetic benchmark and evaluated locally deployed large language models for contextual stigma detection without exposure to protected health information.

METHODS

PIPELINE



SYNTHETIC BENCHMARK DESIGN

500 synthetic clinical sentences were generated: 200 clearly stigmatizing, 200 clearly neutral, and 100 ambiguous "gray" cases designed to evaluate contextual ambiguity beyond keyword matching.

Example: "Patient claims her living situation is stable, though she was unable to provide a current address."

Does 'claims' reflect objective clinical documentation or disbelief here? Expert adjudication established the final benchmark label.

CLINICIAN ADJUDICATION

Reviewed by board-certified Addiction Medicine physician.
Final Benchmark Labels: Stigmatizing (n=251) | Neutral (n=249)
Example adjudication rules:

Term	Stigmatizing When
Refused	Frames reasonable behavior as obstinance
Claims	Implies clinician disbelief
Attribution	Provider-sourced; not lay family
Despite	Judgmental framing rather than chronology
Clean/Dirty	Frames test results using moralized language

MODEL EVALUATION

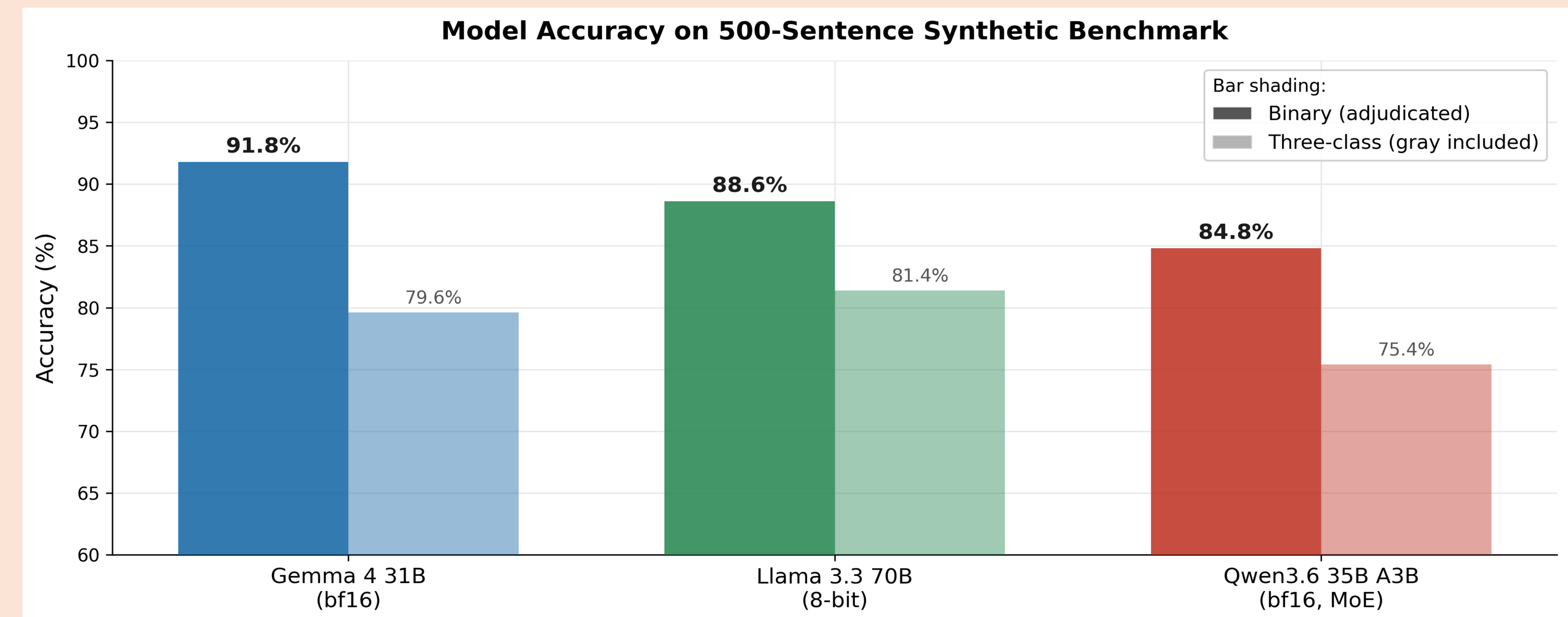
Zero-shot local inference on consumer Apple Silicon (Gemma 4 31B bf16, Llama 3.3 70B 8-bit, Qwen3.6-35B A3B) via MLX framework.

EXTERNAL VALIDATION

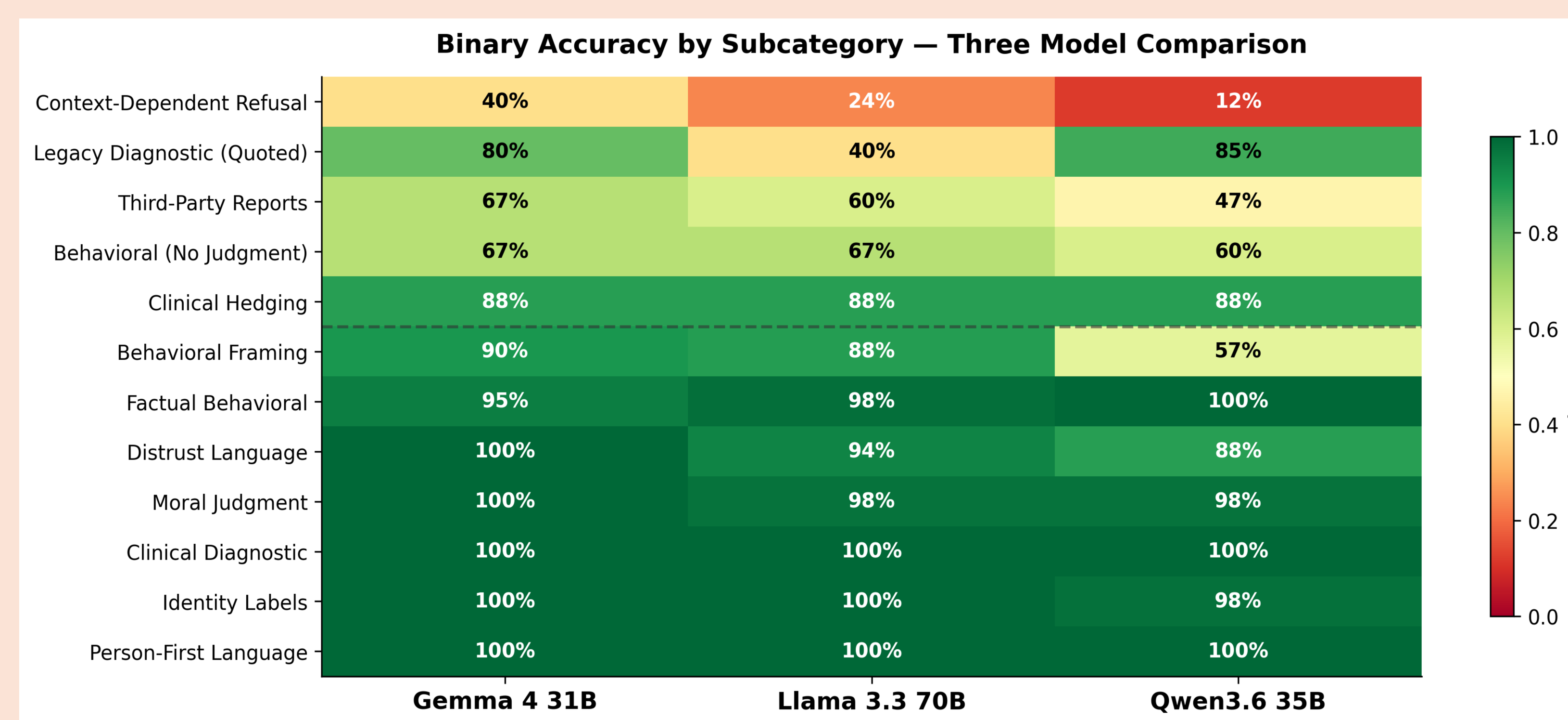
Validated using a SUD-filtered subset of the Harrigian MIMIC-IV[7,8] annotated corpus (n=238).

RESULTS

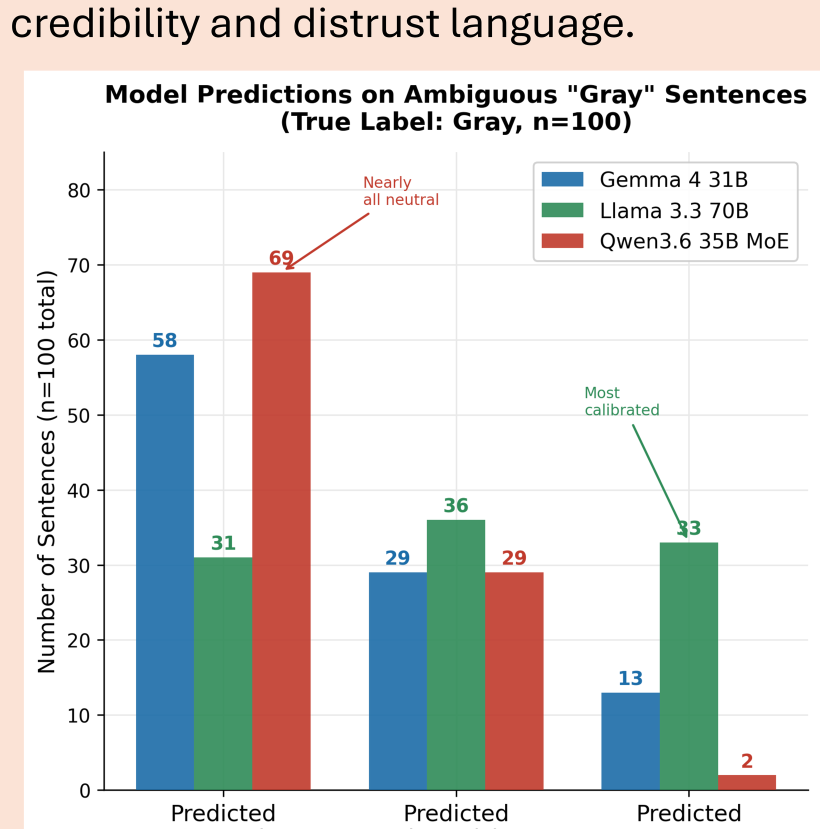
Gemma 4 31B achieved the highest binary accuracy (91.8%). Llama 3.3 70B best preserved uncertainty on ambiguous "gray" cases (gray F1 = 0.423). Qwen3.6 35B A3B achieved 84.8% accuracy but rarely assigned gray classifications (gray F1 = 0.038), suggesting a tendency toward definitive rather than uncertain classifications.

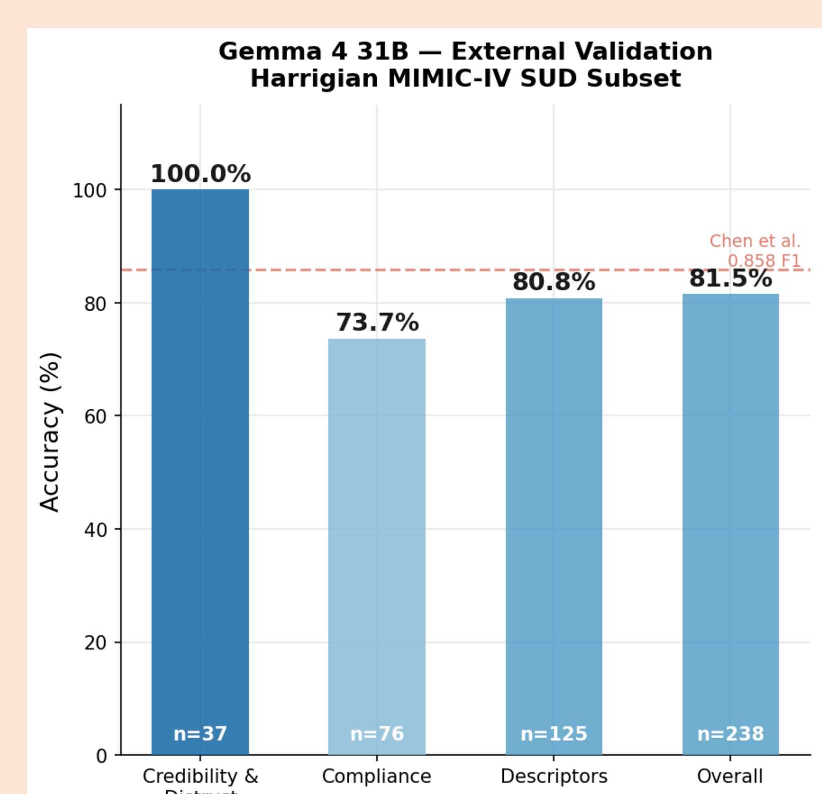


All models achieved 100% accuracy on explicit stigma categories. Errors clustered in adjudicated gray-case scenarios, suggesting that contextual framing, not explicit labeling, remains the core detection challenge.



Models demonstrated distinct behavior under contextual ambiguity. Llama best preserved uncertainty across adjudicated gray cases, whereas Gemma and Qwen favored more definitive classifications. External validation of Gemma 4 31B on a SUD-filtered MIMIC-IV subset demonstrated 81.5% overall accuracy and 100% accuracy for credibility and distrust language.





CONCLUSION

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- Gemma 4 31B achieved the highest binary accuracy (91.8%) on a clinician-reviewed synthetic benchmark, outperforming Llama 3.3 70B (88.6%) and Qwen3.6 35B MoE (84.8%).
- Frontier language models can reliably generate clinically valid stigmatizing and neutral clinical sentences, enabling privacy-preserving benchmark development without requiring access to real patient records.
- Models demonstrated distinct behavior under contextual ambiguity. Llama best preserved uncertainty on adjudicated gray cases (gray F1: 0.423), whereas Gemma and Qwen favored more definitive classifications.
- External validation yielded 100% accuracy on credibility and distrust language and 81.5% overall accuracy on a SUD-filtered MIMIC-IV subset, supporting generalizability to real clinical text.
- Locally deployed LLMs on consumer Apple Silicon can detect both explicit and implicit stigmatizing language without fine-tuning or patient data exposure, preserving privacy.

LIMITATIONS

- Synthetic benchmark sentences may not fully capture the complexity of real clinical notes
- External validation limited to SUD-filtered subset (n=238); sentence boundaries approximated due to annotation offset differences.
- Gray case adjudication reflects a single clinician's framework; inter-rater reliability assessment is ongoing.
- Models evaluated zero-shot; fine-tuning on domain-specific data may improve performance

AUTHORS & DISCLOSURES

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Board-Certified: Addiction Medicine | Clinical Informatics | Emergency Medicine.

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